

AI Development

Track 03 - AI Solutions Engineering

ASSIGNMENT 4

Retrieval-Augmented Knowledge Assistant

Objective

Build a RAG-based chatbot that answers questions from uploaded or predefined documents.

Professional Scope

Implement document ingestion plan, chunking strategy, embeddings/retrieval concept, context-aware response, source reference display, and fallback when answer is not found.

Deliverables

- RAG architecture diagram.
- Working RAG chatbot prototype (deployed).
- Sample knowledge base documents.
- Test questions with evaluation sheet.
- README and AI_USAGE.md.

Quality Bar

Responses must be grounded in the knowledge base and should not invent unsupported facts. Source references should be visible when an answer comes from documents. Fallback must clearly guide the user when the answer is not found.

Suggested Tools

LangChain

LlamaIndex

Chroma

Supabase

Pinecone

Streamlit

React

Mandatory Submission Package

GitHub Repository	Use naming format: ProjectName-InnoVias t. Clean folders, useful commits, and a clear branch history.
README.md	Overview, problem statement, features, tech stack, setup steps, screenshots, deployment link, and learning outcomes.
Live Preview	Vercel, Netlify, Render, Firebase, Expo preview, Streamlit, or equivalent based on project type.
Demo Media	1-2 minute screen recording plus 3-5 high-quality screenshots.
Progress Evidence	Screenshots of design, development, bugs fixed, and final improvements.

■ Working Rules

- You may use AI tools for ideation, debugging, UI suggestions, and documentation, but you must understand and explain your own work.
- Keep the scope clear and small - build a complete feature well instead of an unfinished large one.
- Never commit API keys, passwords, tokens, or private credentials to GitHub.
- Post your progress and demo professionally on LinkedIn / Instagram / Facebook where required by the mentor team.

Evaluation Weighting

Project Functionality	30%
UI/UX Quality	20%
Code Structure	15%
GitHub Management	10%
Creativity & Problem Solving	10%
Presentation & Public Showcase	10%
Consistency & Deadline Discipline	5%